Project Presentation of Co-Op Project at Industry (Module-1)

***AgroVision AI**: Smart Insect Detection & Pest Advisory System*

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Introduction & Objective

The main objective of this project is to develop an AI-powered insect detection system that can identify crop pests in real time and provide accurate pesticide recommendations.

Objective of the Project

- Detect insects and pests from crop images using AI and computer vision.
- Recommend suitable pesticides based on pest type and infestation level.
- Reduce excessive pesticide usage.
- Improve crop productivity and farmer decision-making.
- Promote sustainable and precision-based agriculture.

Methodology

The system collects and preprocesses insect images using image enhancement and data augmentation techniques. A Convolutional Neural Network (CNN) model is trained to identify and classify crop pests accurately. Farmers upload crop images through a mobile or web application for real-time pest detection. The AI model analyzes the image and predicts the insect type along with infestation details. Based on the prediction results, the system recommends suitable pesticides and preventive measures to the farmer.

Key Findings

The AI-based system successfully detects and classifies crop insects with high accuracy using deep learning techniques. Real-time pest identification helps farmers take faster and more effective decisions for crop protection. The pesticide recommendation feature reduces excessive chemical usage and lowers farming costs. The system promotes sustainable agriculture by minimizing environmental damage and improving soil health. Overall, the project demonstrates the potential of AI and computer vision in modern smart farming practices.

Conclusion

The AI-powered insect detection and pesticide recommendation system provides an efficient and smart solution for modern agriculture. By using artificial intelligence and computer vision, the system can accurately identify crop pests and recommend suitable pesticides in real time. This helps farmers reduce crop losses, minimize excessive pesticide usage, and improve overall productivity. The project also supports sustainable and precision-based farming practices.

Future Scope

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- Integration of IoT sensors and drones for automatic field monitoring.
- Support for multiple regional languages and voice assistance.
- Detection of plant diseases and nutrient deficiencies along with insects.
- Integration with weather forecasting for early pest outbreak prediction.
- Development of a complete smart farming platform using cloud and mobile technologies.

References

- TensorFlow Documentation and tutorials for deep learning and CNN models.
- OpenCV for image processing and insect detection techniques.
- PyTorch resources for AI model training and classification.
- Research papers on AI-based pest detection from IEEE journals and conferences.
- Agricultural pest datasets and information from Food and Agriculture Organization resources.

Thank You